SQLite: More Powerful Than You Think

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SQLite

- Embedded database
- A single file, or in-memory
- Library ≈650KB
- Since 2000
- Public domain
- 3 developers
- 1 trillion databases worldwide: browsers, phones, planes, cars, TVs, cameras…

Demo: SQLite

Application-Defined SQL Functions

- Simple functions
- Window functions
- Aggregate functions
- Table-valued functions

Vector Extensions

Local Vector Search

- **sqlite-vec**:
  - in alpha
  - sponsored by Mozilla, Turso, and others
  - Last release in 2024
- **sqlite-vector**:
  - led by SQLite AI, highly optimized
  - Last release yesterday

Session Extension

- Open a session
- Modify the database
- Create a changeset file
- Apply the generated file to another database to synchronize it
- Can be reversed (A→A' but also A'→A)
- Changesets can be merged
- Even supports "rebase"

sqlite-http

- by Alex Garcia
- pre-v1
- no observability
- good for rapid prototyping (LLM)

zipfile

Read/modify files inside an archive

Write the rows of a query result into an archive, one file per row

Use cases:

- Find a class/string in a JAR or docx
- Modify data in zip files in a one-time script
- Transform your query results into an archive

Other Capabilities/Extensions

Built into SQLite:

- SQLite Archive
- Spell Checking

Extensions:

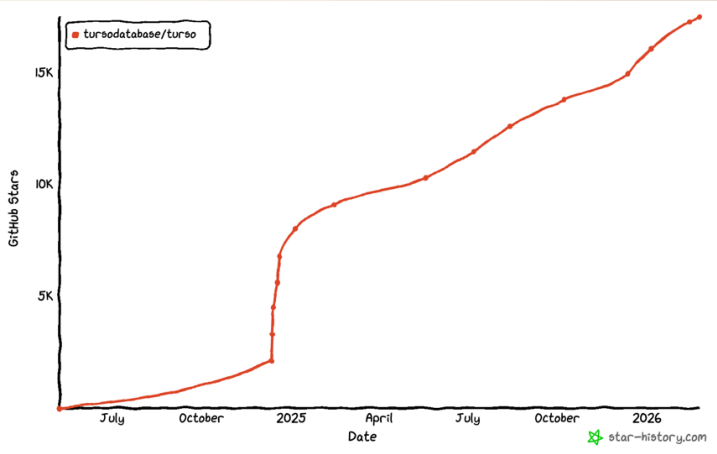
- sqlelf
- sqlite-js
- litesql/kafka
- litesql/nats

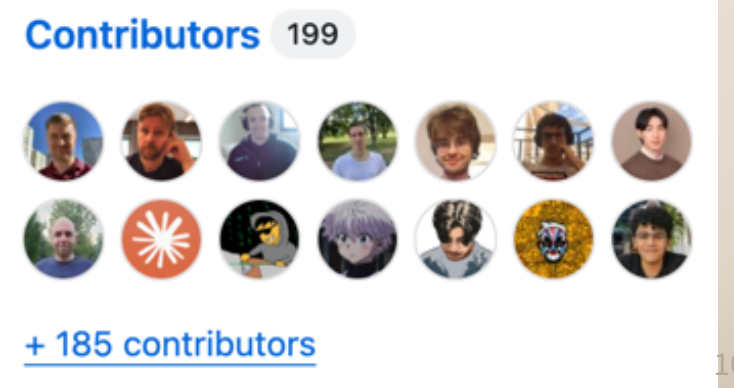
Replication Tools

- libSQL

Turso

- SQLite-compatible
- Async-first
- Encryption, Vector Search
- Multi-Writer
- MIT license
- 17K GH stars
- 200 contributors





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Link to the code examples

https://github.com/LeMikaelF/confoo-2026-sqlite